# Criterion D: Development


Criterion D (12 marks) is the core of your IA, where you show your product and coding skills. It's not enough to just paste code. You must use *code excerpts* to explain and *justify* why you chose specific algorithms and techniques. You must also provide *evidence* (like test tables) that you executed your test plan from Criterion C. Your 5-minute video is the key to proving your product is "fully functional."

## Criterion D: Building and Documenting Your Solution

This is the most important criterion, worth **12 marks**, and it's where the majority of your 2,000-word count will go.

**Goal:** To demonstrate your programming skills, justify your development choices, and provide clear evidence that your product is fully functional and well-tested.

**Recommended Word Count:** ~1,000 words

Criterion D is marked on three key pillars:

1. **Functionality:** Does your product work? (Proven by the video).
2. **Appropriate Techniques:** How did you build it, and why? (Proven by your code excerpts and justifications).
3. **Effectiveness of Testing:** How do you *know* it works? (Proven by your test evidence).

### Pillar 1: Functionality

A "fully functional product" is one that successfully solves the problem you defined back in Criterion A.

- **How to Prove It:** This is the **primary purpose of your 5-minute video**. The video must showcase all the main features of your product working.
- A product that doesn't match your problem specification from Criterion A (e.g., you said you'd build a database app, but you made a simple website) cannot achieve full marks.
- Check out our **The 5-Minute IA Video: A Practical Guide** for a full breakdown.

### Pillar 2: Appropriate Techniques (The Core Write-up)

This is where you show you are a *computer scientist*, not just a *coder*. You must demonstrate your algorithmic thinking by explaining the techniques you used to turn your algorithms from Criterion C into working code.

**Justify, Don't Just Paste**

The biggest mistake students make is pasting large blocks of code with no explanation. The goal is to **justify your choices**.

**How to use code excerpts correctly:**

1. **Use Excerpts:** Select small, important sections of your code that demonstrate a specific technique (e.g., a sorting algorithm, a database query, an OOP class definition, handling an API call).
2. **Include Comments:** Your code should be well-commented.
3. **Explain:** Briefly describe *what* the code does, providing details and explaining how it works.
4. **Justify & Evaluate:** This is the most important part. Explain *why* you chose this technique.
    - *Good Justification:* "I chose to implement a **binary search** for the library's lookup feature. Because the book IDs are already sorted, this provides **O(log n)** time complexity, which is significantly more efficient than a linear search's O(n) time as the database grows."
    - *Bad Justification:* "This is my search function. It uses a loop to find the book."
5. **Reference the Appendix:** Every code excerpt must reference the full source code, which **must** be in your appendix. (e.g., "See `Appendix A: main.py`, lines 90-115").

## Pillar 3: Effectiveness of Testing (The Evidence)

Here, you provide the *evidence* that you actually performed the tests you *planned* in Criterion C.

**Tool for the Job: The "Observed Results" Table**

The best way to show this is to copy your **Test Plan table from Criterion C** and add two new columns: "Observed Outcome" and "Pass/Fail."

- You must provide evidence for **at least three** comprehensive test cases.
- This must include **functional testing** (testing your Success Criteria) and **structural testing** (testing with **valid, invalid, and extreme** data).
- Your **video** should also show you performing some of these tests (especially the invalid data tests!).

| Test ID | Test Data | Expected Outcome | Observed Outcome | Pass/Fail |
|---|---|---|---|---|
| T-01 | "abc" | An error message "Invalid input: Please enter a number." is displayed. | An error message "Invalid input: Please enter a number." was displayed. | **Pass** |

| | | | | |
|---|---|---|---|---|
| T-02 | User: "admin", Pass: "1234" | The user is logged in and taken to the main menu. | The user was logged in and the main menu appeared. | **Pass** |
| T-03 | User: "admin", Pass: "wrong" | An error message "Login failed: Incorrect username or password." is displayed. | An error message "Login failed: Incorrect username or password." was displayed. | **Pass** |
| T-04 | 99999999 | The item is added to the database successfully. | The program crashed with an "Integer Overflow" error. | **Fail** |

A "Fail" is not a bad thing! It shows your testing was effective. In your write-up, you would then explain *why* it failed and how you fixed it (or would fix it, if you ran out of time).

## Using AI Ethically

The IB and the Teacher Support Material are clear: **Yes, you can use AI tools** to help you debug or refine code, but you **MUST** acknowledge it.

If you use an LLM:

1. **In your code:** Add a comment explaining what the AI helped with.
2. **In your documentation:** Add a note in Criterion D explaining how you used the tool.

Example Acknowledgement (from TSM):

"I wrote a function to sort a list, but it had a logic error. ChatGPT suggested using Python's sorted() function, improving efficiency from $O(n^2)$ to $O(n \log n)$. I've acknowledged this change in my comments."

## 🛑 Final Reminder: The Appendix

Your Criterion D write-up will only have *excerpts* of code. Your **full, complete, and commented source code** must be included as an appendix. If it's not there, you cannot get the highest marks.

Report a problem